KLE Society's

KLE Technological University, Hubballli

Department Of MCA

MCA IV Semester

Synopsis Report

On

"Automated Single-Image to 3D Asset and Point Cloud Generation System"

Submitted by

Rajashekhar B Durgad

01FE24MCA027

Under the Guidance of

Prof. Akash Hulkund

KLE Technological University

Vidyanagar, Hubballi – 580031

2025-2026

TABLE OF CONTENTS

1. INTRODUCTION

With the rapid advancement of computer vision, deep learning, and 3D graphics technologies, industries such as e-commerce, augmented reality (AR), robotics, and industrial inspection face a growing demand for intelligent systems capable of automating the digital reconstruction of physical objects from images. Traditional 3D digitization workflows rely on manual modeling, specialized photogrammetry equipment, or controlled multi-view capture setups that are time-intensive, costly, and skill-dependent.

The project **"Automated Single-Image to 3D Asset and Point Cloud Generation System"** is a fully integrated, end-to-end AI pipeline that takes a single photograph as input and automatically produces a high-fidelity textured 3D mesh and a semantically segmented 3D point cloud of the primary object within it. The system was developed to address the need for rapid 3D content creation for AR/VR product visualization workflows, eliminating manual modeling bottlenecks.

The pipeline integrates three state-of-the-art foundation models:

- **GroundingDINO** for zero-shot open-vocabulary object detection.
- **Florence-2** for image captioning, prompt generation, and part-level detection.
- **SAM2.1** for precise instance segmentation.

These models form a **Universal Object Detection Pipeline** capable of handling diverse image conditions (including dark, low-contrast, grayscale, and crowded scenes). The detected primary object is automatically cropped, background-removed, and passed to the **Hunyuan3D-2** generative framework, which produces high-fidelity textured 3D meshes in GLB format. The system then converts the mesh into a dense 3D point cloud and segments it geometrically using DBSCAN clustering, providing a complete 3D representation and analysis workflow.

2. PROJECT OBJECTIVES

The primary objectives of this project are:

- **Universal Object Detection Pipeline:** To design and implement a detection pipeline combining Grounding DINO, Florence-2, and SAM2.1 with adaptive CLAHE image enhancement and multi-pass fallback strategies, enabling robust detection under challenging conditions (dark, low-contrast, and grayscale images).
- **Prompt-Free Operation:** To develop automated image captioning and prompt generation using Florence-2 to allow fully prompt-free operation on arbitrary input images without manual text inputs.
- **High-Fidelity 3D Generation:** To integrate Hunyuan3D-2 for single-image to textured 3D mesh generation via background removal, RGBA crop extraction, shape generation, and appearance-flow UV texture synthesis.

- **Interactive 3D Visualization:** To implement an interactive auto-rotating 3D mesh and point cloud viewer using the Open3D library.
- **Mesh to Point Cloud Conversion:** To build a conversion utility using Poisson-disk surface sampling and surface normal estimation.
- **Point Cloud Geometric Segmentation:** To implement a point cloud segmentation tool supporting DBSCAN clustering, RANSAC plane fitting, and voxel-based decomposition with color-labeled cluster visualization.
- **Validation:** To validate the complete pipeline end-to-end on real-world objects across all pipeline stages.

3. PROBLEM STATEMENT

- **Fragmentation:** No unified automated workflow connects object detection, background removal, 3D mesh generation, and point cloud analysis. Each stage typically requires a separate tool and manual data transfer between stages.
- **Brittleness under challenging imaging conditions:** Most detection pipelines assume controlled lighting and high-contrast scenes, and fail on dark, low-contrast, or grayscale images that are common in real-world product photography.
- **High barrier to entry:** Producing a single high-quality 3D model requires several hours of skilled manual modeling or dozens of calibrated photographs in photogrammetry pipelines, creating significant time and cost bottlenecks.

Therefore, there is a need for an intelligent, robust, and extensible single-image-to-3D asset pipeline that leverages state-of-the-art vision-language models and generative frameworks to automate 3D reconstruction and downstream point cloud segmentation in a fully prompt-free, end-to-end manner.

4. SCOPE OF THE PROJECT

- **Input Handling:** Support for standard image formats (JPEG, PNG, WebP, BMP) with automatic adaptive preprocessing for dark, low-contrast, and grayscale images.
- **Object Detection & Segmentation:** Zero-shot whole-object detection, part-level detection, and pixel-precise instance segmentation using foundation models.
- **3D Asset Generation:** Automated production of watertight 3D meshes with high-fidelity PBR textures in GLB format from a single input photograph.
- **Point Cloud Processing:** Conversion of generated 3D meshes to dense point clouds (target: 100,000 points), surface normal estimation, and multi-method geometric segmentation.
- **Output Compatibility:** Exporting GLB files compatible with Blender, Three.js, and WebXR platforms, and PLY point cloud files compatible with Open3D, MeshLab, and CloudCompare.

5. LITERATURE REVIEW

5.1 Existing Systems Overview

Existing 3D digitization systems focus on reconstructing physical objects into digital 3D representations using photogrammetry, manual modeling, or AI-based single-image reconstruction. Most systems demonstrate functionalities such as object detection, mesh generation, and texture mapping using deep learning frameworks or cloud-based AI services. While these solutions enhance productivity and enable new applications in AR/VR, they often depend heavily on controlled imaging conditions or fragmented tool chains. As a result, challenges related to robustness, automation, and end-to-end integration remain unresolved.

5.2 Limitations of Existing Work

- The system provides zero-shot detection but assumes controlled lighting and high-contrast scenes [1]
- Unified vision-language capabilities but lacks 3D reconstruction and geometric modeling [2]
- Pixel-precise segmentation but only produces 2D masks without 3D mesh generation [3]
- High-quality textured 3D mesh generation but requires clean background-free RGBA input [4]
- Fast feedforward 3D reconstruction but produces lower texture quality and lacks built-in segmentation [7]

5.3 Comparative Analysis of Reviewed Papers

Paper Name	Technologies Used	Limitations
GroundingDINO: Open-Set Object Detection [1]	DINO Visual Encoder, BERT Text Encoder, Transformer	Assumes controlled lighting; fails on dark/low-contrast inputs without preprocessing
Florence-2: Unified Vision Tasks [2]	Vision-Language Sequence-to-Sequence Model	Lacks 3D reconstruction and geometric modeling capabilities
SAM 2: Segment Anything [3]	Promptable Visual Segmenter, Hiera Backbone	Only generates 2D masks; lacks 3D mesh generation capabilities
Hunyuan3D-2: 3D Asset Generation [4]	Flow-Matching DiT, Appearance-Flow UV Synthesis	Requires clean, background-free RGBA input; high VRAM utilization
TripoSR: Fast 3D Reconstruction [7]	Large Reconstruction Model (LRM), NeRF/Meshes	Lower texture quality; lacks built-in object segmentation

6. PROPOSED SYSTEM

The proposed system is a fully integrated, end-to-end automated pipeline that accepts a single input image and produces a textured 3D mesh and a segmented 3D point cloud without manual intervention. The pipeline is organized into four sequential macro-stages:

- **Stage 1 — Input and Enhancement:** Computes mean brightness and contrast standard deviation. Applies adaptive CLAHE enhancement in the LAB color space when the image is dark (brightness < 0.30) or low-contrast (contrast std $<$ 0.15).
- **Stage 2 — Universal Object Detection:** Florence-2 generates a natural-language caption, converted to a GroundingDINO-compatible prompt via noun extraction. GroundingDINO performs zero-shot detection with a multi-pass adaptive threshold strategy ($0.20 \rightarrow 0.15 \rightarrow 0.10$). Florence-2 identifies object parts, and SAM2.1 generates pixel-precise binary masks.
- **Stage 3 — 3D Generation:** Background is removed using rembg with ONNX GPU runtime, creating an RGBA crop. This crop is passed to Hunyuan3D-2 Stage 1 (flow-matching shape generation) and Stage 2 (appearance-flow UV texture synthesis at 1024-pixel resolution) to produce a PBR-textured GLB mesh.
- **Stage 4 — Point Cloud Processing:** Converts the GLB mesh to a dense point cloud (100,000 points) via Poisson-disk sampling, estimates normals, and segments the cloud using DBSCAN Euclidean clustering, exporting a color-labeled PLY file.

6.1 Applications of the Proposed System

1. Desktop Automation and E-commerce

- Rapid automated 3D product asset generation from catalog photographs for interactive 3D viewing
- Reduces manual modeling and photography effort

2. Augmented Reality & VR

- Direct asset export (GLB) for WebXR, Three.js, and Unity/Unreal prototyping
- Enables immersive product visualization experiences

3. Robotics and Industrial Inspection

- Segmented point clouds providing geometric part decomposition for robotic grasp planning
- Generating 3D digital twins of physical components for dimensional comparison

7. SYSTEM ARCHITECTURE

The architecture integrates deep learning models and geometric algorithms in a sequential workflow. To prevent Out-Of-Memory (OOM) errors on a single 16 GB VRAM GPU, the system implements a strict memory management strategy: each model is loaded sequentially, performs inference, and is completely unloaded from memory (with explicit CUDA cache clearing) before the next model is loaded.

7.1 Architecture Flow

User → Input Image → Adaptive Enhancement (CLAHE) → Florence-2 Auto-Caption → GroundingDINO Detection → Florence-2 Part Detection → SAM2.1 Segmentation → rembg Background Removal → Hunyuan3D-2 Shape Generation → Hunyuan3D-2 Texture Synthesis → GLB Mesh Output → Poisson-Disk Point Cloud Conversion → DBSCAN Segmentation → Color-Labeled PLY Output

Architecture Diagram

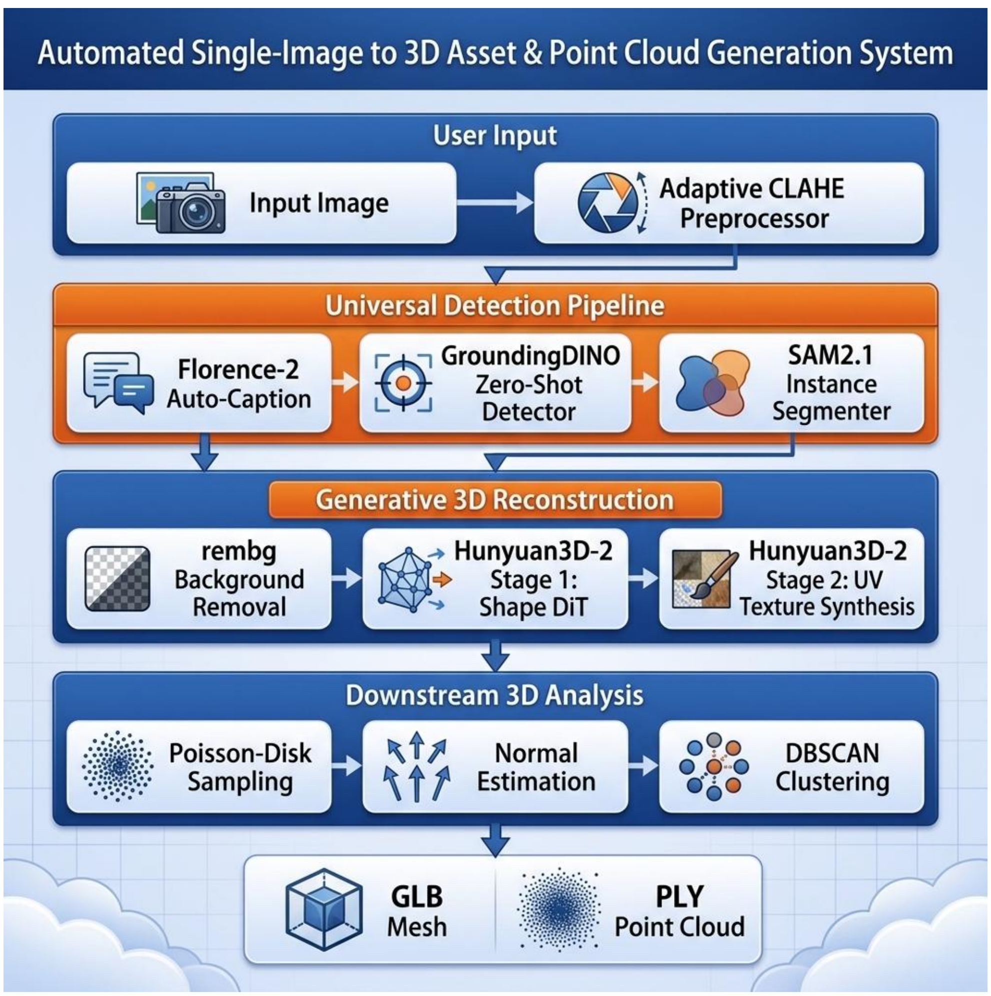

Figure 1: System Architecture Diagram showing the four macro-stages and data flow

Block Diagram

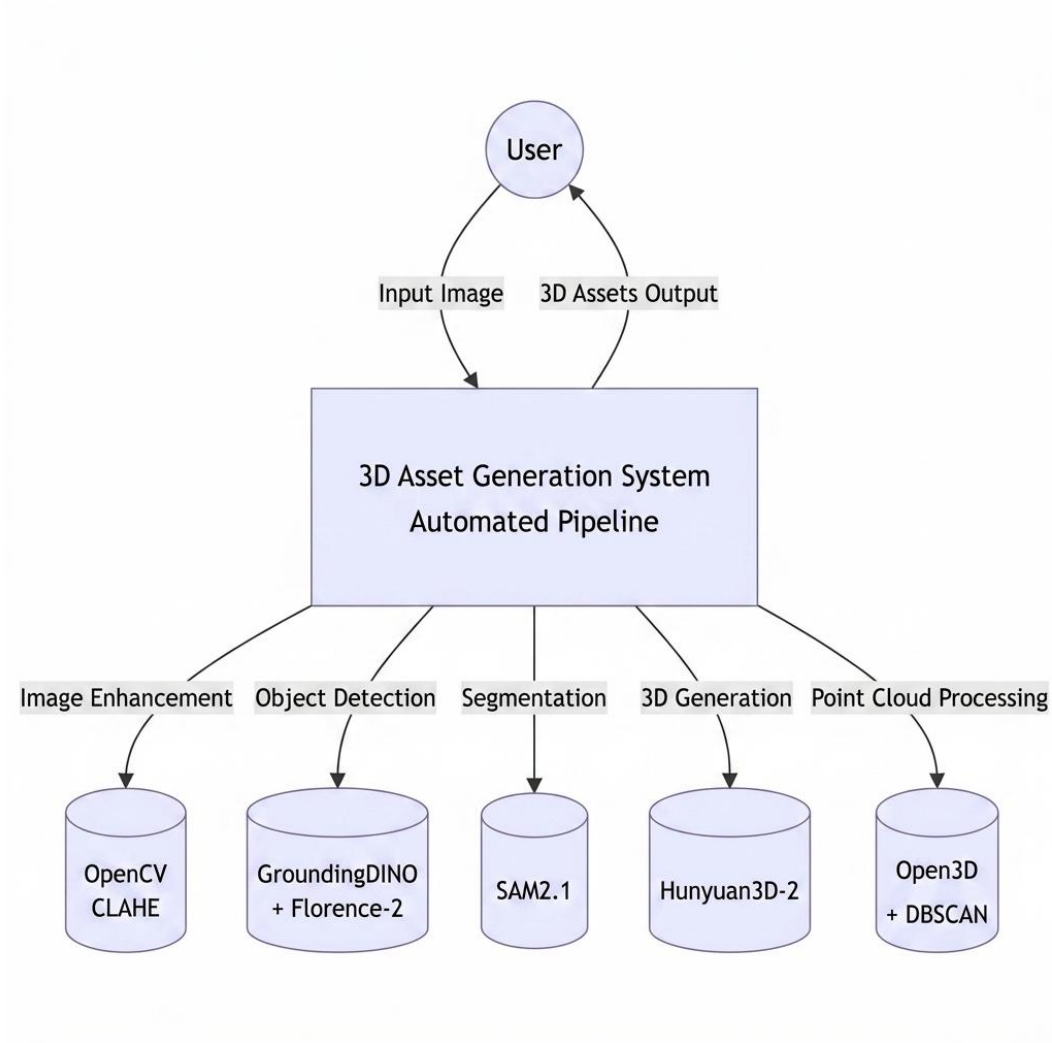

Figure 2: System Block Diagram showing control and data flow

8. MODULES OF THE PROJECT

- **Speech Recognition / Image Loading Module:** Loads the input image in any standard format and performs adaptive CLAHE enhancement in the LAB color space if the image is dark or low-contrast.
- **NLP / Auto-Caption Module (Florence-2):** Runs image captioning in float16 precision and extracts content nouns to build the GroundingDINO text prompt, enabling fully prompt-free operation.
- **Detection Module (GroundingDINO):** Executes multi-pass zero-shot object detection with threshold relaxation (0.20 → 0.15 → 0.10) to find bounding boxes under challenging conditions.

- **Segmentation Module (Florence-2 + SAM2.1):** Identifies sub-component bounding boxes and generates high-quality instance masks with IoU-based selection.
- **3D Generation Module (Hunyuan3D-2):** Produces watertight mesh geometry via Stage 1 Shape Diffusion and bakes PBR UV texture mapping via Stage 2 Texture Synthesis, saving a GLB file.
- **Visualization Module:** Opens an interactive, auto-rotating 3D rendering window (1280×720 at 6 RPM) using Open3D.
- **Point Cloud Conversion Module:** Converts GLB to PLY format using Poisson-disk surface sampling (100,000 points) and estimates consistently oriented surface normals.
- **Segmentation and Export Module:** Removes statistical outliers and segments the point cloud into distinct spatial clusters using DBSCAN, coloring each cluster according to a fixed palette.

9. SOFTWARE AND HARDWARE REQUIREMENTS

9.1 Software Requirements

- **Operating System:** Linux (Ubuntu 20.04+) or Windows
- **Programming Language:** Python 3.10 or later
- **Deep Learning Framework:** PyTorch 2.x with CUDA 12.1
- **3D Processing Libraries:** Open3D ≥ 0.16, trimesh
- **Image Processing Libraries:** OpenCV, Pillow, NumPy, Matplotlib
- **Background Removal:** rembg with ONNX GPU runtime
- **Model Hub:** HuggingFace Transformers

9.2 Hardware Requirements

- **Processor:** Intel Core i5 or AMD Ryzen 5 (or higher)
- **RAM:** Minimum 12 GB system RAM (16 GB recommended)
- **GPU:** NVIDIA GPU with minimum 16 GB VRAM (e.g., NVIDIA T4)
- **Storage:** Minimum 50 GB free disk space
- **Input Device:** Microphone-enabled system (optional for future voice expansion)
- **Output Device:** Monitor with 1280×720 or higher resolution

10. METHODOLOGY

- Requirement analysis and literature study
- System architecture and pipeline design
- Adaptive preprocessing and detection implementation
- 3D mesh reconstruction pipeline integration
- Point cloud processing suite development
- Interactive visualization development
- End-to-end testing and validation
- Documentation

11. EXPECTED OUTCOMES

- Accurate detection and segmentation of objects from single images
- High-fidelity textured 3D mesh generation in GLB format
- Dense point cloud generation with surface normals
- Automated geometric segmentation of point clouds
- Reduced manual effort in 3D content creation
- Robust operation under challenging image conditions

12. ADVANTAGES OF THE PROPOSED SYSTEM

- **Prompt-Free Operation:** Florence-2 removes the need for manual text tagging, enabling full automation.
- **Robustness:** Adaptive CLAHE and multi-pass detection ensure high reliability on imperfect images.
- **Scalability:** Supports seamless addition of new detection models or 3D generators without major system changes.

13. FUTURE SCOPE

- **Real-Time Optimization:** Implementing TensorRT compilation and INT8 quantization to reduce latency to under 1 minute.
- **Multi-Object Reconstruction:** Extending the bounding box pipeline to process multiple objects and reconstruct complex 3D environments.
- **Video Integration:** Utilizing SAM2.1's video tracking capability to generate 3D models from rotating product video clips.
- **Web Interface:** Developing a Gradio or Streamlit-based web dashboard for drag-and-drop image uploading and asset downloading.

14. REFERENCES

[1] S. Liu et al. "GroundingDINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection," ECCV 2024.

[2] B. Xiao et al. "Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks," CVPR 2024.

[3] N. Ravi et al. "SAM 2: Segment Anything in Images and Videos," arXiv:2408.00714, 2024.

[4] Tencent Hunyuan3D Team. "Hunyuan3D-2: Scaling Diffusion Models for High Resolution Textured 3D Assets Generation," arXiv:2501.12202, 2025.

[5] Q.-Y. Zhou, J. Park, and V. Koltun. "Open3D: A Modern Library for 3D Data Processing," arXiv:1801.09847, 2018.

[6] M. Ester et al. "A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise," KDD 1996.

[7] D. Tochilkin et al. "TripoSR: Fast 3D Object Reconstruction from a Single Image," arXiv:2403.02156, 2024.